Tushar Anand

+91 7488441023 tushar.annandwork@gmail.com linkedin.com/in/tusharannand github.com/neoandmatrix

Experience

AsyncApi Mentee **Remote**

Open Source Contributor *July 2026 - September 2026*

- Contributed to the **AsyncAPI** projects sponsored by **RedHat and IBM**. Added new commands, enhancing existing ones, and improving test coverage for a reliable developmet. Cleaned up unused dependencies and reduced project build size across platforms.
- Implemented a new **preview command** enhancing the end user UX and additon of hedless browser testing with puppeteer for studio command in cli, for which a reward of $400 was awarded under the AsyncApi bounty program.
- Upgraded dependencies while **maintaining compatibility** across AsyncAPI templates to improve performance and maintainability along with improving build times by 40% and reducing docker size with multi stage builds.
- Contributed to the **AsyncApi Studio** poject by adding support for **local schema file** previewing with CLI.
- Got a total of **26 pull requests** merged across CLI and other repositories along with helping maintainers in review.

Clickpe (YC W23) **Remote**

AI Engineer Intern *January 2026 - July 2026*

- Built **AI based voice calling** infrastructure to **handle 5k+** incoming and outgoing calls to customers on daily basis.
- Orchestrated an **Agentic pipeline** of **stt**,**llm** and **tts** models using **pipecat** with **sub 200ms** latency during calls.
- Reduced **tts** cost by **30%** using redis to **cache audio blobs** with hashing and **llm** costs by **60%** with **prefix caching**.
- Integrated **langfuse** for tracing along with **custom dashboards** to track various metrics and **detect hallucinaitons**.
- Created **n8n workflows** to automatically **schedule calls** and internal dashboard to **track status** of ongoing calls.
- Used **in-memory rag** to i**nsert dynamic information** in prompt based on user query during the call **reducing** system prompt **size by 40%.** and improving **inference** speed **by 10-15% on average**.

Creuto **Remote**

Software Development Intern *Jan 2025 - July 2025*

- Developed the backend for an internal application from scratch for **scheduling meetings** within the company, reducing the need for 3rd party applications and cost by **70%**.
- Developed a **microservice based payment architecture** for **Razorpay** integration with existing infrastructure.
- Make My Look: Developed the backend for a salon booking application incorporating **Razorpay** for payments,**Firebase cloud messaging** for on device reltime notificaions and **MongoDB geospatial** queries for **location based service** recommendation and **redis** for caching popular services data handling over **1k+ users efficiently**.

Projects

Anymemory - The Second you | *Typescript, Cloudflare,Vectorize,Llama*

- Built a **persistent information storage engine** based on **Hierarchical Memory Decomposition** to store and retrive information across sources like **pdfs,urls,text,images** etc. Used **vector db** for storing **raw embedding**.
- Created **waypoints** for related memories. Used sql database for **meatdata** and **R2 object storage** for **multimedia**.
- Leveraged **vision models** to generate image metadata, LLMs to classify memories into five cognitive sectors, and **hybrid BM25 + vector** search with **waypoint mean embeddings** for semantic retrieval.
- Implemented an **MCP server** enabling AI agents to retrieve and interact with stored memories.

GoJudge – Distributed Online Judge | *Golang,docker,Typescript*

- Built an **online judge** for university competitive programming contests featuring contest creation, **multi-language code execution**, automated judging, **WebSocket**-based live leaderboards, and **real-time** contest updates.
- Architected a **fault-tolerant** asynchronous execution pipeline using **Redis** and **BullMQ**, decoupling APIs from **distributed judge** workers to reliably process concurrent submissions with **automatic retry and recovery**.
- Executed untrusted code inside a secure **gVisor sandbox**, enforcing CPU, memory, filesystem, and network isolation while defending against resource exhaustion, privilege escalation, sandbox escape, and denial-of-service attacks.
- Instrumented the platform with **Prometheus** metrics and **Grafana** dashboards to monitor submission throughput, queue depth, worker utilization, execution latency, and sandbox failures for operational visibility.

Technical Skills

Languages:C, Typescript, Python, SQL, Bash
Technologies: Git, Docker, Cloudflare, AWS, Postgres, Github Actions, Kubernetes, Linux

Education

International Institute of Information Technology **Bhubaneshwar, Odisha**

Bachelor of Technology in ***Computer Engineering*** ***Aug 2023 - Aug 2027***

GPA: 8.1/10.0

- **Coursework:** Data Structures And Algorithms, Statics, Computer Architecture and Digital Logic, Networking.